



Guide – Domain name server

Objectives:

- DNS configuration
- Zone creation
- Direct and reverse zones

1. Install domain name server

```
apt-get update  
apt-get install bind9
```

2. Create a new zone with the name lr.estga.ua.pt com in which your server is the master server. Edit /etc/bind/named.conf.local and add the following lines at the bottom of the file:

```
zone "lr.estga.ua.pt" IN {  
    type master;  
    file "/etc/bind/zones/lr.estga.ua.pt";  
    allow-query{any;};  
};
```

3. Create the zone file (/etc/bind/zones/lr.estga.ua.pt)

```
mkdir /etc/bind/zones/  
vi /etc/bind/zones/lr.estga.ua.pt
```

4. Editing the file and place there the following lines:

```
$ORIGIN lr.estga.ua.pt.  
$TTL      604800  
@         IN      SOA      ns1r.lr.estga.ua.pt. email.do.gestor. (  
                                1              ; Serial  
                                7200            ; Refresh  
                                120            ; Retry  
                                2419200        ; Expire  
                                604800)        ; Default TTL  
@         IN      NS       ns1r.lr.estga.ua.pt.  
ns1r      IN      A        192.168.1.1
```

5. Validate server's configuration and zone file.

```
named-checkconf  
named-checkzone lr.estga.ua.pt /etc/bind/zones/lr.estga.ua.pt
```



Note: Validation output reports problems that have to be fixed before proceeding.

6. Restart the server

```
/etc/init.d/bind9 restart
```

7. Test server's resolution requesting `ns1r.lr.estga.ua.pt` resolution.

```
nslookup ns1r.lr.estga.ua.pt 127.0.0.1
```

8. Add A records for zone machines:

- Hostmachine - 192.168.1.1
- Debapp- 192.168.1.2
- Win2008 - 192.168.1.3
- Elastix - 192.168.1.4
- Zentyal - 192.168.1.5
- pasg - 192.168.1.6
- lmat - 192.168.1.254

9. Test server's resolution for the records added.

Note: Changes in configuration files require a server restart to take effect.

10. Add a MX record with a 10 value to the domain email server named `mail.lr.estga.ua.pt` at the bottom of SOA record.

```
@                IN          MX          10    mail.lr.estga.ua.pt.
```

Note: MX record just points out email server name, but it does not resolves its name.

11. Test server MX record

```
nslookup -query=MX lr.estga.ua.pt
```

12. Create an alias to `mail.lr.estga.ua.pt` named `www`, adding the following line to the zone records:

```
www      IN      CNAME  mail.lr.estga.ua.pt
```

13. Test `www.lr.estga.ua.pt` resolution.

14. Create a new reverse zone named `1.168.192.IN-ADDR.ARPA` in which your server is the a master server. Edit `/etc/bind/named.conf.local` and add the following lines at the bottom of the file:

```
zone "1.168.192.IN-ADDR.ARPA" IN {  
    type master;  
    file "/etc/bind/zones/192.168.1.rev";  
};
```



15. Edit reverse zone file (/etc/bind/zones/192.168.1.rev) and add there a SOA record accordingly with the following lines:

```
@      IN      SOA      ns1r.l1r.estga.ua.pt. admin.email.addr. (
                                2011101501      ; Serial
                                7200      ; Refresh
                                120      ; Retry
                                2419200      ; Expire
                                604800)      ; Default TTL

      IN      NS      ns1r.l1r.estga.ua.pt.
1      IN      PTR      ns1r.l1r.estga.ua.pt.
```

16. Validate servers and reverse zone configuration.

17. Test server reverse resolution:

```
nslookup 192.168.1.1 127.0.0.1
```

18. Taking in consideration the information from 8., complete reverse zone information.

19. Test servers resolution of names belonging to other zones:

```
nslookup mail.ua.pt 127.0.0.1
```

20. Configure forwarders addresses to the server, adding to the bottom of the file /etc/bind/named.conf.options the following lines:

```
forwarders {
    193.136.172.20;
    193.136.172.21;
};
```

21. Repeat the resolution of names belonging to other zones:

```
nslookup mail.ua.pt 127.0.0.1
```

1.

```
ines@Ubuntu22:~$ apt-get update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
ines@Ubuntu22:~$ sudo apt-get update
[sudo] password for ines:
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:1 https://pt.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 https://pt.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 https://pt.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu jammy InRelease
Reading package lists... Done
```

2.

```
ines@Ubuntu22: ~
GNU nano 6.2 /etc/bind/named.conf.local
//
// Do any local configuration here
//

zone "lr.estga.ua.pt" IN {
    type master;
    file "/etc/bind/zones/lr.estga.ua.pt";
    allow-query{any;};
};
```

3.

```
ines@Ubuntu22:~$ sudo mkdir /etc/bind/zones/
ines@Ubuntu22:~$ sudo nano /etc/bind/zones/lr.estga.ua.pt
```

4.

```
ines@Ubuntu22: ~
GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt *
$ORIGIN lr.estga.ua.pt.
$TTL      604800
@         IN      SOA      ns1r.l1r.estga.ua.pt. email.do.gestor. (
                                1          ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

@         IN      NS       ns1r.l1r.estga.ua.pt.
ns1r      IN      A        192.168.1.1
```

5.

```
ines@Ubuntu22:~$ named-checkconf
ines@Ubuntu22:~$ named-checkzone lr.estga.ua.pt /etc/bind/zones/lr.estga.ua.pt
zone lr.estga.ua.pt/IN: loaded serial 1
OK
```

6.

```
ines@Ubuntu22:~$ sudo systemctl restart named
```

(nome do serviço do bind no Ubuntu é 'named')

comandos úteis:

```
>> sudo systemctl start named
```

```
>> sudo systemctl status named
```

```
>> sudo systemctl restart named
```

```
>> sudo systemctl stop named
```

```
ines@Ubuntu22:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset:▶
   Active: active (running) since Fri 2025-11-07 11:56:23 GMT; 38min ago
     Docs: man:named(8)
   Process: 6497 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUC▶
  Main PID: 6498 (named)
    Tasks: 23 (limit: 28055)
   Memory: 26.1M
      CPU: 100ms
   CGroup: /system.slice/named.service
           └─6498 /usr/sbin/named -u bind

Nov 07 11:56:23 Ubuntu22 named[6498]: managed-keys-zone: loaded serial 0
Nov 07 11:56:23 Ubuntu22 named[6498]: zone 0.in-addr.arpa/IN: loaded serial 1
Nov 07 11:56:23 Ubuntu22 named[6498]: zone 255.in-addr.arpa/IN: loaded serial 1
Nov 07 11:56:23 Ubuntu22 named[6498]: zone 127.in-addr.arpa/IN: loaded serial 1
Nov 07 11:56:23 Ubuntu22 named[6498]: zone localhost/IN: loaded serial 2
Nov 07 11:56:23 Ubuntu22 named[6498]: all zones loaded
Nov 07 11:56:23 Ubuntu22 named[6498]: running
Nov 07 11:56:23 Ubuntu22 systemd[1]: Started BIND Domain Name Server.
Nov 07 11:56:33 Ubuntu22 named[6498]: managed-keys-zone: Unable to fetch DNSKEY▶
Nov 07 11:56:33 Ubuntu22 named[6498]: resolver priming query complete: timed out
lines 1-22/22 (END)...skipping...
```

7.

```
ines@Ubuntu22:~$ nslookup ns1r.lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   ns1r.lr.estga.ua.pt
Address: 192.168.1.1
```

Como configurado previamente no exercício 4, a resolução acima está correta.

8.

```
GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt *
$ORIGIN lr.estga.ua.pt.

$TTL      604800
@         IN      SOA      ns1r.l1r.estga.ua.pt. email.do.gestor. (
                                1          ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

@         IN      NS       ns1r.l1r.estga.ua.pt.
ns1r      IN      A        192.168.1.1
Hostmachine IN      A        192.168.1.1
Debapp    IN      A        192.168.1.2
Win2008   IN      A        192.168.1.3
Elastix   IN      A        192.168.1.4
Zentyal   IN      A        192.168.1.5
pasg      IN      A        192.168.1.6
lmat      IN      A        192.168.1.254
```

9.

```
ines@Ubuntu22:~$ sudo nano /etc/bind/zones/lr.estga.ua.pt
ines@Ubuntu22:~$ named-checkconf
ines@Ubuntu22:~$ named-checkzone lr.estga.ua.pt /etc/bind/zones/lr.estga.ua.pt
zone lr.estga.ua.pt/IN: loaded serial 1
OK
ines@Ubuntu22:~$ sudo systemctl restart named
ines@Ubuntu22:~$ nslookup Hostmachine.l1r.estga.ua.pt 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   Hostmachine.l1r.estga.ua.pt
Address: 192.168.1.1

ines@Ubuntu22:~$ nslookup Debapp.l1r.estga.ua.pt 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   Debapp.l1r.estga.ua.pt
Address: 192.168.1.2

ines@Ubuntu22:~$ nslookup Win2008.l1r.estga.ua.pt 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   Win2008.l1r.estga.ua.pt
Address: 192.168.1.3

ines@Ubuntu22:~$ nslookup Elastix.l1r.estga.ua.pt 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   Elastix.l1r.estga.ua.pt
Address: 192.168.1.4
```

```

ines@Ubuntu22:~$ nslookup Zentyal.lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   Zentyal.lr.estga.ua.pt
Address: 192.168.1.5

ines@Ubuntu22:~$ nslookup pasg.lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   pasg.lr.estga.ua.pt
Address: 192.168.1.6

ines@Ubuntu22:~$ nslookup lmat.lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   lmat.lr.estga.ua.pt
Address: 192.168.1.254

```

10.

```

GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt *
$ORIGIN lr.estga.ua.pt.

$TTL      604800
@         IN      SOA      ns1r.lr.estga.ua.pt. email.do.gestor. (
                        1          ; Serial
                        7200       ; Refresh
                        120        ; Retry
                        2419200    ; Expire
                        604800)    ; Default TTL

@         IN      MX       10 mail.lr.estga.ua.pt.
@         IN      NS       ns1r.lr.estga.ua.pt.
ns1r      IN      A        192.168.1.1
Hostmachine IN      A        192.168.1.1
Debapp    IN      A        192.168.1.2
Win2008   IN      A        192.168.1.3
Elastix   IN      A        192.168.1.4
Zentyal   IN      A        192.168.1.5
pasg      IN      A        192.168.1.6
lmat      IN      A        192.168.1.254
mail      IN      A        192.168.1.4

```

Coloquei um registo A para resolver o nome do email server mail.lr.estga.ua.pt para, por exemplo, o endereço 192.168.1.4 (mas podia ter escolhido um dos outros endereços).

11.

```

ines@Ubuntu22:~$ nslookup -query=MX lr.estga.ua.pt
Server:          127.0.0.53
Address:         127.0.0.53#53

** server can't find lr.estga.ua.pt: NXDOMAIN

```



```
ines@Ubuntu22:~$ nslookup -query=MX lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

lr.estga.ua.pt  mail exchanger = 10 mail.lr.estga.ua.pt.
```

12.

```
GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt *
$ORIGIN lr.estga.ua.pt.

$TTL      604800
@         IN      SOA     ns1r.lr.estga.ua.pt. email.do.gestor. (
                                1          ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

@         IN      NS      ns1r.lr.estga.ua.pt.
@         IN      MX      10 mail.lr.estga.ua.pt.
ns1r      IN      A        192.168.1.1
Hostmachine IN      A        192.168.1.1
Debapp    IN      A        192.168.1.2
Win2008   IN      A        192.168.1.3
Elastix   IN      A        192.168.1.4
Zentyal   IN      A        192.168.1.5
pasg      IN      A        192.168.1.6
lmat      IN      A        192.168.1.254
mail      IN      A        192.168.1.4
www       IN      CNAME    mail.lr.estga.ua.pt
```

```
www       IN      CNAME    mail.lr.estga.ua.pt
```

13.

```
ines@Ubuntu22:~$ nslookup -query=MX www.lr.estga.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

** server can't find www.lr.estga.ua.pt: NXDOMAIN
```


14.

```
GNU nano 6.2 /etc/bind/named.conf.local *
//
// Do any local configuration here
//

zone "lr.estga.ua.pt" IN {
    type master;
    file "/etc/bind/zones/lr.estga.ua.pt";
    allow-query{any;};
};

zone "1.168.192.IN-ADDR.ARPA" IN {
    type master;
    file "/etc/bind/zones/192.168.1.rev";
};
```

15.

```
GNU nano 6.2 /etc/bind/zones/192.168.1.rev *
@      IN      SOA      ns1r.lr.estga.ua.pt. admin.email.addr. (
                                2011101501      ; Serial
                                7200             ; Refresh
                                120              ; Retry
                                2419200          ; Expire
                                604800)          ; Default TTL

@      IN      NS       ns1r.lr.estga.ua.pt.
1      IN      PTR      ns1r.lr.estga.ua.pt.
```

16.

```
ines@Ubuntu22:~$ named-checkzone lr.estga.ua.pt /etc/bind/zones/192.168.1.rev
/etc/bind/zones/192.168.1.rev:1: no TTL specified; using SOA MINTTL instead
zone lr.estga.ua.pt/IN: NS 'ns1r.lr.estga.ua.pt' has no address records (A or AAAA)
zone lr.estga.ua.pt/IN: not loaded due to errors.
```

17.

```
ines@Ubuntu22:~$ nslookup 192.168.1.1 127.0.0.1
1.1.168.192.IN-ADDR.ARPA      name = ns1r.1.168.192.IN-ADDR.ARPA.
1.1.168.192.IN-ADDR.ARPA      name = Hostmachine.1.168.192.IN-ADDR.ARPA.
1.1.168.192.IN-ADDR.ARPA      name = ns1r.lr.estga.ua.pt.
```

18.

```
GNU nano 6.2 /etc/bind/zones/192.168.1.rev *
@      IN      SOA      ns1r.lr.estga.ua.pt. admin.email.addr. (
                                2011101501      ; Serial
                                7200             ; Refresh
                                120              ; Retry
                                2419200          ; Expire
                                604800)          ; Default TTL

1      IN      NS       ns1r.lr.estga.ua.pt.
1      IN      PTR      ns1r.lr.estga.ua.pt.
1      IN      PTR      Hostmachine
2      IN      PTR      ns1r
3      IN      PTR      Debapp
4      IN      PTR      Win2008
5      IN      PTR      Elastix
6      IN      PTR      Zentyal
4      IN      PTR      pasg
254    IN      PTR      mail.lr.estga.ua.pt
254    IN      PTR      lmat
```

19.

```
ines@Ubuntu22:~$ nslookup mail.ua.pt 127.0.0.1
;; communications error to 127.0.0.1#53: timed out
;; Got SERVFAIL reply from 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

** server can't find mail.ua.pt: SERVFAIL
```

Não funciona, pois mail.ua.pt não é um nome cuja tradução exista nesta zona (sem ser reverse). Nessa primeira zona, existe a configuração para mail.lr.estga.ua.pt com registo A, com nome mapeado para o endereço 192.168.1.4

Na zona reverse, há a correspondência oposta, com o endereço 192.168.1.4 mapeado para o nome mail.lr.estga.ua.pt

Assim, alterando, já funciona:

```
ines@Ubuntu22:~$ nslookup mail.lr.estga.ua.pt 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   mail.lr.estga.ua.pt
Address: 192.168.1.4
```

20.

```
GNU nano 6.2 /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        193.136.172.20;
        193.136.172.21;
    };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.lsc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};
```

21.

```
ines@Ubuntu22:~$ sudo systemctl restart named
Job for named.service failed because the control process exited with error code.
See "systemctl status named.service" and "journalctl -xeu named.service" for details
.
```

```
ines@Ubuntu22:~$ nslookup mail.ua.pt 127.0.0.1
;; communications error to 127.0.0.1#53: connection refused
;; communications error to 127.0.0.1#53: connection refused
;; communications error to 127.0.0.1#53: connection refused
;; no servers could be reached
```

Na rede Wi-Fi de casa não foi possível porque não consegue contactar os DNS Server Forwarders.

Fazendo na ESTGA, funciona como pretendido:

```
ines@Ubuntu22:~$ sudo systemctl restart named
ines@Ubuntu22:~$ nslookup mail.ua.pt 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Non-authoritative answer:
Name:   mail.ua.pt
Address: 193.136.173.7
```